Kubernetes Taints and Tolerations

Taints and tolerations are Kubernetes mechanisms used to **control which Pods can be scheduled onto which Nodes**.

The easiest mental model is:

Taint = "Keep Pods away from this Node."
Toleration = "This Pod is allowed to stay on a tainted Node."

They work together, but they are **not the same thing**.

1. Why do we need Taints and Tolerations?

Imagine you have a cluster with:

Node 1 → General applications

Node 2 → Database workloads

Node 3 → GPU workloads

Node 4 → Production-critical workloads

You don't want a normal application Pod accidentally landing on the GPU or database node.

For example:

For example:

GPU Node

|

| "Don't schedule normal Pods here"

|

X

You can put a **taint** on the GPU node.

Only Pods that explicitly have the appropriate **toleration** can be scheduled there.

Taint

GPU Node ----------> "No normal Pods"

GPU Pod ---- toleration ----> Allowed

Normal Pod ----------> Rejected

2. What is a Taint?

A **taint is applied to a Node**.

It tells Kubernetes:

"Pods that don't tolerate this taint should not be placed here."

Conceptually:

Node

└── Taint

  ├── key

  ├── value

  └── effect

Example:

key=workload

value=gpu

effect=NoSchedule

Meaning:

Only Pods that tolerate workload=gpu can be scheduled onto this node.

3. What is a Toleration?

A **toleration is applied to a Pod**.

It tells Kubernetes:

"This Pod can tolerate a particular taint."

Example:

tolerations:

```
- key: workload
  operator: Equal
  value: gpu
  effect: NoSchedule
```

This does **not** mean:

"Put this Pod on the GPU node."

It only means:

"This Pod is allowed to be scheduled on a node having this taint."

This distinction is extremely important.

4. Taint vs Toleration

Feature	Taint	Toleration
Applied to	Node	Pod
Purpose	Repel Pods	Allow Pod to tolerate taint
Says	"Keep out"	"I'm allowed"
Used by	Scheduler / kubelet behavior	Scheduler / kubelet behavior
Automatically attracts Pod?	No	No

Interview point

Toleration does not guarantee scheduling onto the tainted node.

You may need **nodeSelector**, **node affinity**, or another placement mechanism if you specifically want the Pod to run there.

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5. Taint Structure

A taint has three important components:

key=value:effect

For example:

environment=production:NoSchedule

Here:

Key = environment

Value = production

Effect = NoSchedule

Another example:

workload=gpu:NoSchedule

6. Three Taint Effects

Kubernetes has three important taint effects:

NoSchedule

PreferNoSchedule

NoExecute

Understanding these is very important for interviews.

6.1 NoSchedule

effect=NoSchedule

Meaning:

Pods that don't tolerate this taint will not be newly scheduled onto the node.

Example:

Node:

gpu=true:NoSchedule

A Pod without the matching toleration:

❌ Cannot be newly scheduled

A Pod with the toleration:

✅ Can be scheduled

Important

Existing Pods that are already running on the node are generally **not evicted just because the NoSchedule taint is added**.

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7. PreferNoSchedule

effect=PreferNoSchedule

This is a **soft restriction**.

Kubernetes tries to avoid placing Pods on the node if they don't tolerate the taint, but it is not an absolute prohibition.

Think:

NoSchedule

↓

"DO NOT schedule here"

PreferNoSchedule

↓

"Prefer NOT to schedule here"

It is therefore less strict than NoSchedule.

8. NoExecute

This is the strongest effect.

effect=NoExecute

It affects both:

1. **New Pods**
2. **Existing Pods**

If a Pod does not tolerate the NoExecute taint, it can be **evicted from the node**.

Example:

Node gets:

maintenance=true:NoExecute

Existing Pod:

Pod A

❌ doesn't tolerate it

↓

Evicted

Existing Pod with matching toleration:

Pod B

✅ tolerates it

↓

Can remain

9. The Three Effects — Easy Memory Trick

Remember:

NoSchedule

↓

New Pods → blocked

PreferNoSchedule

↓

New Pods → preferably avoided

NoExecute

↓

New Pods → blocked

Existing Pods → can be evicted

Interview question

Which taint effect can evict existing Pods?

Answer:

NoExecute

10. How Taints and Tolerations Work Together

Suppose Node has:

dedicated=database:NoSchedule

A normal Pod:

No toleration

Result:

❌ Pod cannot be scheduled there

A database Pod:

tolerations:

- key: dedicated

  operator: Equal

  value: database

  effect: NoSchedule

Result:

✅ Pod is allowed there

But remember:

Toleration ≠ Node selection

It only removes the taint-based restriction.

11. Toleration Operators

There are two important operators you'll commonly see.

Equal

Example:

```
tolerations:
  - key: dedicated
    operator: Equal
    value: database
    effect: NoSchedule
```

This means:

key must match

AND

value must match

AND

effect must match

Conceptually:

dedicated=database:NoSchedule

Exists

Example:

```
tolerations:
  - key: dedicated
    operator: Exists
    effect: NoSchedule
```

This means:

The Pod tolerates the taint as long as the specified key exists, regardless of its value.

For example, it can tolerate:

dedicated=database:NoSchedule

and potentially:

dedicated=gpu:NoSchedule

because the key is:

dedicated

Important

With:

operator: Exists

you normally **do not specify value**.

12. Toleration with No Effect

You can also omit the effect in some toleration configurations.

For example:

```
tolerations:
  - key: dedicated
    operator: Exists
```

This can match taints with that key across the relevant effects according to Kubernetes toleration matching rules.

For interviews, remember the basic explicit form first:

key

operator

value

effect

13. NoExecute and tolerationSeconds

NoExecute has another important feature:

---

tolerationSeconds

It allows a Pod to tolerate a NoExecute taint for a limited amount of time.

Example:

```
tolerations:
  - key: maintenance
    operator: Equal
    value: planned
    effect: NoExecute
    tolerationSeconds: 60
```

Meaning approximately:

Node gets maintenance=planned:NoExecute

↓

Pod tolerates it

↓

60 seconds

↓

Pod can be evicted

This is useful when you don't want immediate eviction.

14. Multiple Taints

A Node can have multiple taints.

For example:

Node

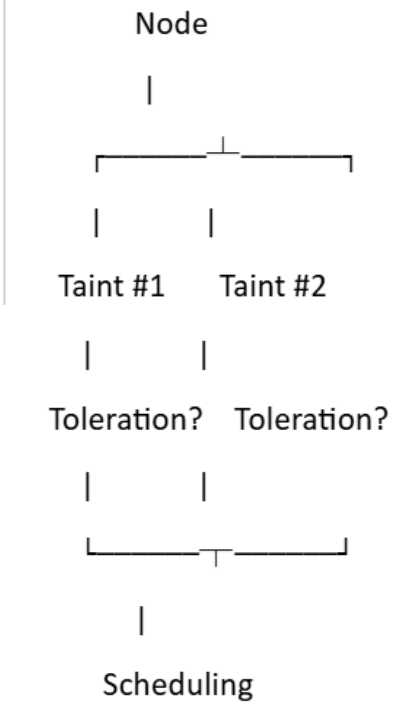
├— environment=production:NoSchedule

├— workload=gpu:NoSchedule

└— maintenance=false:NoExecute

A Pod must tolerate the relevant taints to remain eligible.

Think of it as multiple doors:



If an applicable NoSchedule taint isn't tolerated, the Pod cannot be scheduled there.

15. Taints Do NOT Attract Pods

This is one of the most common misunderstandings.

Suppose:

Node A:

workload=gpu:NoSchedule

And your Pod has:

```
tolerations:
 - key: workload
   operator: Equal
   value: gpu
   effect: NoSchedule
```

Some people think:

"The Pod will automatically go to Node A."

❌ Not necessarily.

The toleration only says:

"Node A is permitted."

The scheduler still considers other scheduling requirements and constraints.

16. Taints + NodeSelector

If you want:

"Only GPU Pods should run on GPU nodes."

A common pattern is:

Taint

+

Node label

+

Toleration

+

NodeSelector/Affinity

For example:

GPU Node

17. Taints vs NodeSelector

These solve different problems.

NodeSelector

Says:

"I want my Pod on nodes having this label."

Taint

Says:

"I don't want arbitrary Pods on this node."

Therefore:

NodeSelector

↓

Pod → Node

Taint/Toleration

↓

Node ← Pod

A useful way to remember:

A useful way to remember:

NodeSelector attracts Pods toward matching nodes.
Taints repel Pods from nodes unless tolerated.

18. Real-World Use Cases

1. GPU nodes

GPU nodes

↓

taint

↓

Only GPU workloads allowed

2. Dedicated database nodes

Database nodes

↓

taint

↓

Only database workloads tolerate it

3. Production workloads

You can dedicate certain nodes to critical workloads.

Production node

↓

NoSchedule

↓

Only approved workloads

4. Maintenance

A node can be marked with a NoExecute taint so workloads that don't tolerate it are removed.

5. Special hardware

Examples:

GPU

High-memory

SSD

ARM

Special NIC

Taints can prevent normal workloads from consuming specialized resources.

20. Taints vs Pod Affinity

These are also different.

Taint/Toleration

Controls:

"Which Pods are allowed on this node?"

Node Affinity

Controls:

"Which nodes does this Pod prefer/require?"

Pod Affinity

Controls:

"Which Pods should be located near other Pods?"

Pod Anti-Affinity

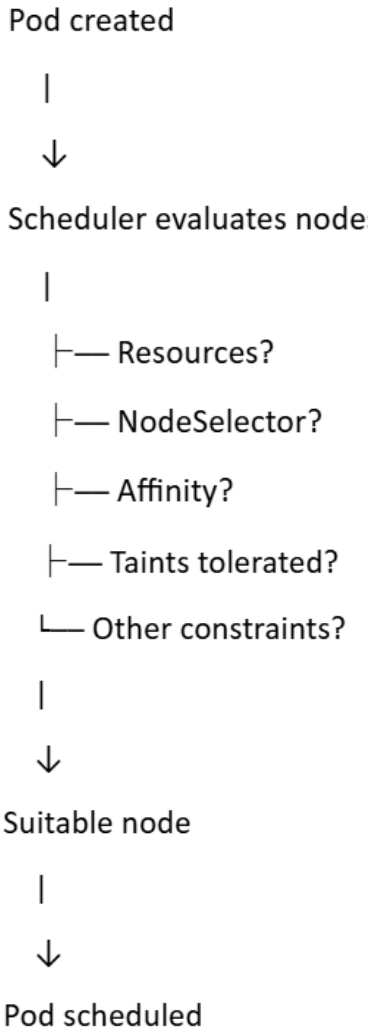
Controls:

"Which Pods should be separated from other Pods?"

21. Important Kubernetes Architecture Point

Taints are primarily a **node-side scheduling constraint**, while tolerations are Pod specifications that make certain taints acceptable.

A simplified scheduling flow:



22. Taints and Eviction

Be particularly careful with:

NoExecute

It is not merely a scheduling rule.

It can affect Pods **already running** on the node.

For example:

Node

|

| NoExecute taint

↓

Existing Pods

|

├— Pod A → no toleration → eviction

├— Pod B → toleration → remains

└— Pod C → tolerationSeconds → remains temporarily

This is why NoExecute is especially important during node problems and maintenance.

23. Built-in Kubernetes Taints

Kubernetes can automatically add certain taints under node conditions.

For example, node problems can lead to taints associated with conditions such as:

node.kubernetes.io/not-ready

node.kubernetes.io/unreachable

node.kubernetes.io/memory-pressure

node.kubernetes.io/disk-pressure

node.kubernetes.io/pid-pressure

node.kubernetes.io/network-unavailable

24. Master/Control-Plane Nodes

Historically, Kubernetes control-plane nodes commonly have a taint similar to:

node-role.kubernetes.io/control-plane:NoSchedule

This prevents ordinary workload Pods from being scheduled there.

Conceptually:

Control Plane

|

| NoSchedule

↓

Normal workload ❌

> Control-plane-tolerating Pod
>
> ↓
>
> Possible ✅
>
> This is why when you have a small lab cluster, you may sometimes need to understand control-plane taints when trying to schedule workloads.

25. Most Important Interview Questions

Q1. What is a taint?

A taint is applied to a node to repel Pods that don't have a matching toleration.

Q2. What is a toleration?

A toleration is applied to a Pod and allows it to be scheduled onto or remain on a node with a matching taint.

Q3. Does toleration force a Pod onto a node?

No. It only allows the Pod to tolerate the taint.

Q4. Which taint effect evicts existing Pods?

NoExecute.

Q5. Difference between NoSchedule and NoExecute?

NoSchedule

→ blocks new scheduling

NoExecute

→ blocks new scheduling + can evict existing Pods

Q6. What is PreferNoSchedule?

A soft preference to avoid scheduling Pods that don't tolerate the taint.

Q7. Where is taint configured?

On the Node.

Q8. Where is toleration configured?

In the Pod specification.

Q9. How do you ensure a Pod actually goes to a tainted node?

Typically combine:

Taint

+

Toleration

+

NodeSelector / Node Affinity